



FITTING WIND SPEED DISTRIBUTIONS: A CASE STUDY

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Abstract—The wind speed data represented in the form of frequency curves show the shape of a potential model. The Weibull and Lognormal models are used for this purpose, with hourly mean wind speed data. This study deals with the estimation of the annual Weibull and Lognormal parameters from 20 locations in Navarre. The suitability of both distributions is judged from the R^2 coefficient with a linear regression for the Weibull distribution and a nonlinear regression for the Lognormal distribution. Both approaches give a good fit, giving better results for the Weibull distribution. A comparison between the estimation and the production for a wind farm is offered. © 1998 Elsevier Science Ltd. All rights reserved.

1. INTRODUCTION

The Energy Plan of Navarre, that tries to increase the level of energetic self-sufficiency by means of the development of renewable energy sources, provides a remarkable stimulus to the use of wind energy. In a first approach, wind has been considered a possible power capacity of around 2400 MW, although the location of a few potential windfarms is not possible in areas with a special environmental interest. For this reason, the forecasts reckon around 636 MW from now till 2020. That would imply a mean production of 1469 GWh, and it would avoid the emission of approximately 1.3 Mton of CO₂ per year if other conventional energy sources were used. The best location for the installation of wind energy conversion systems must be selected according to a few basic criteria such as a high mean wind speed, acceptable daily and seasonal variations, and both limited turbulence levels and extreme winds. For this reason, statistical studies are important to know the probabilities that some circumstances happen, e.g. wind speed over a value.

One of the numeric methods that can be used is the treatment of historical data sets from meteorological stations to simulate the behaviour of the wind. As an energy source, the natural wind is extremely variable. During the last few years there has been considerable interest in finding a statistical model of wind speed frequency distribution in order to predict the energy output of a wind energy conversion system (WECS). Information about real time

performance is also important. Two different functions have been used to model the wind features of Navarre: the Weibull distribution and the Lognormal distribution. Both models have been chosen because of their frequent use in the literature.

2. WIND DATA

2.1. Meteorological observation network of Navarre

We must first refer to the meteorological network of the Autonomic Community of Navarre, which has been completed and modernized in recent years thanks to collaboration between the Agriculture and Industry Department of the Government of Navarre and the National Institute of Meteorology (Servicio de Estructuras Agrarias, 1995). One of the most important results of this effort has been the set-up of a network of automatic meteorological stations, which is able to supply instantaneous meteorological information. This network started collecting data experimentally in 1992 and, at present, it consists of 25 stations distributed along the community. Some of them are located in highly exposed areas. The expansion and improvement of the network with regard to site density and sensors availability is being studied.

The location of these sites are shown in the enclosed map. Figure 1 distinguishes two types of stations: wind register stations, and others with a higher number of variables, such as

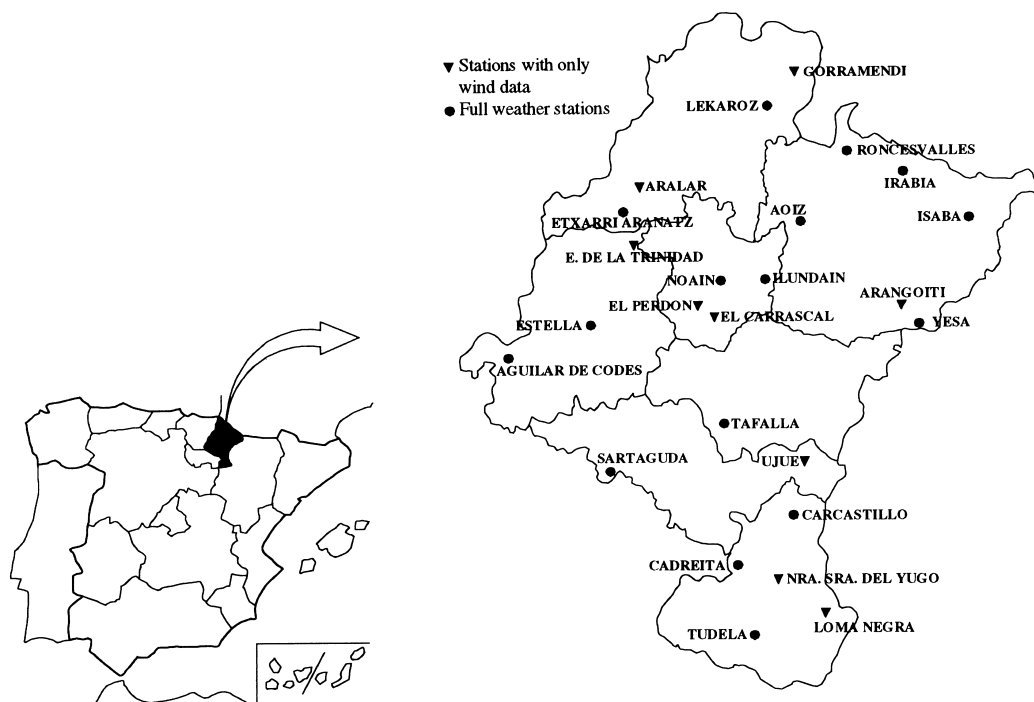


Fig. 1. Location map of the stations.

temperature, rainfall, air pressure, relative humidity, and irradiation.

Information from the meteorological stations of Navarre is incorporated into the Telecommunication World System through the National Institute of Meteorology. In addition, airports, ships, satellites and radars, among others, add information to the system. Prediction models are developed and analyzed from these data.

2.2. Data base

We should first turn our attention for a moment to the data. The received data are stored as daily binary files. The first register of a file corresponds to the first 10 min of the hour 0.00 of that day, and from that point, the program records one register every 10 min. We obtained data from 1992 to 1995.

The following changes were made:

- (1) obtaining monthly files for each station and year, with four columns: month, day, hour and hourly mean wind speed, because the preferred resolution of the series is 1 h. The hourly mean wind speed is the average of the six data corresponding to the periods of 10 min that make up one hour in the original data;
- (2) the lost values were also erased.

After the ASCII files were obtained with these

modifications, they were retrieved in a spreadsheet. We used it for the calculation of the parameters and the adjustment test. The data had already been revised with the incidences document in order to erase the errors that are difficult to detect. When a sensor goes bad, either there are no data or they are incongruous. It may also happen that a damaged sensor produces results with a value that could be true but it is not. These data have been removed.

One year was considered the basic unit for obtaining a set of annual parameters. We defined a complete year as having at least 80% of valid data. The results for 20 stations are given in Table 1. The other ones did not get enough information for making a valid study.

3. THE MODELS

Important efforts have been made since the 1970s to construct an adequate statistical model for describing the wind speed frequency distribution (Stevens and Smulders, 1979; Rehman *et al.*, 1994; Rohatgi *et al.*, 1989). The studied situations give the probability of a wind speed v according to the following expressions for the probability density function:

- (1) Weibull, with c being the scale parameter (value close to the mean speed) and k the

Table 1. Number of hourly average wind speed data

Station	No. of observations				Total
	1992	1993	1994	1995	
Aguilar	8689	8760	8751	8757	34957
Aoiz	8698	8754	8757	8739	34948
Aralar	8011	7144	8665	8158	31978
Arangoiti	8146	8758	7766	-	24670
Cadreita	8003	-	-	8305	16308
Carcastillo	8679	8741	8694	8729	34843
Carrascal	8735	8760	8624	8088	34207
Estella	7265	-	-	-	7265
Etxarri	8433	-	-	-	8433
Gorramendi	7575	8677	8442	-	24694
Ilundain	-	-	7201	8414	15615
Loma Negra	8197	8759	8760	8170	33886
Noain	-	-	-	8648	8648
El Perdón	8759	8759	-	-	-
Sartaguda	-	-	7100	8311	15411
Tafalla	8748	-	-	-	8748
Trinidad	8336	8454	8758	7684	33232
Ujué	8729	8749	8476	8757	34711
Esa	8714	-	-	-	8714
Yugo	8721	8760	-	-	17481

shape parameter:

$$P(v) = (k/c)(v/c)^{k-1} \exp[-(v/c)^k]; \quad (1)$$

- (2) lognormal, where σ and μ are, respectively, the standard deviation and the expected value of the variable $\ln(v)$:

$$P(v) = [(2\pi)^{1/2} \sigma v]^{-1} \times \exp[-(\ln v - \mu)^2 / (2\sigma^2)]; \quad (2)$$

The properties of these distributions are as follows:

- (a) the probability of the wind speed being over a given value v_x , (from the cumulative distribution function) is, for the Weibull distribution:

$$P(v > v_x) = \exp[-(v_x/c)^k], \quad (3)$$

but for the lognormal, because its basis is that $\ln(v)$ is normally distributed, the probability $P(v < v_x)$ must be evaluated from tables of the unit normal distribution, using the standard form with $(\ln v - \mu)/\sigma$;

- (b) the probability of having a wind speed between two values of interest v_x and v_y is, for the Weibull:

$$P(v_x < v < v_y) = \exp[-(v_x/c)^k] - \exp[-(v_y/c)^k], \quad (4)$$

and for the lognormal, it is obtained from tables of unit normal distribution;

- (c) the mean speeds, for the Weibull, with Γ being the gamma function:

$$\bar{v} = c\Gamma[1 + (1/k)], \quad (5)$$

and for the lognormal:

$$\bar{v} = \exp[\mu + (\sigma^2/2)]; \quad (6)$$

- (d) the standard deviation, σ , of the distributions is obtained from the variance; for the Weibull:

$$\sigma = c[\Gamma(1 + 2/k) - \Gamma^2(1 + 1/k)]^{1/2}, \quad (7)$$

and for the lognormal:

$$\sigma = [e^{2\mu} e^{\sigma^2} (e^{\sigma^2} - 1)]^{1/2}. \quad (8)$$

For the two parameter Weibull distribution, numerous papers have appeared over the years dealing with exact or approximate methods of inference to calculate the shape parameter k and the scale parameter c . Several tables and graphs relating to different methods of inference exist. One of these methods is the Weibull probability paper which is based in a transformation of the cumulative distribution function (Johnson *et al.*, 1994). The data are summarized for each station in amplitude classes of 1 m s^{-1} and with a logarithmic transformation of the cumulative distribution function (all the operations have been made with the help of a spreadsheet):

$$P(v < v_i) = P_i; \quad \ln(1 - P_i) = -(v_i/c)^k;$$

$$\ln[-\ln(1 - P_i)] = k \ln(v_i) - k \ln c, \quad (9)$$

setting $\ln[-\ln(1 - P_i)]$ as y and $\ln(v_i)$ as x we obtain the linear relationship $y = kx - k \ln c$. A straight line is observed. The slope of the line is k and the intercept with the y -axis is $-k \ln c$. We used a statistical software package to obtain the parameters of the Lognormal distribution

Table 2. Numerical values of Weibull and Lognormal parameters for annual wind speed distribution and R^2 coefficient

Station	Lognormal			Weibull		
	Mean μ	Standard deviation σ	R^2	Shape k	Scale c	R^2
Aguilar	1.565	0.559	0.9409	1.863	4.864	0.9994
Aoiz	0.929	0.704	0.9560	1.458	2.402	0.9932
Aralar	2.054	0.472	0.9357	2.022	8.095	0.9973
Arangoiti	2.043	0.609	0.9018	1.966	7.696	0.9943
Cadreita	0.945	0.876	0.9557	1.381	2.737	0.9786
Carcastillo	1.125	0.645	0.9901	1.622	3.219	0.9985
Carrascal	1.868	0.669	0.9529	1.906	6.970	0.9974
Estella	0.714	0.529	0.9991	1.354	1.869	0.9859
Etxarri	0.940	0.723	0.9444	1.208	2.589	0.9921
Gorramendi	2.060	0.617	0.9843	1.920	9.486	0.9933
Ilundain	0.573	0.634	0.9990	1.136	1.667	0.9941
Lomanegra	1.911	0.732	0.9719	1.732	7.866	0.9941
Noain	0.922	1.134	0.8464	1.361	3.171	0.9415
Perdon	2.181	0.638	0.9225	2.022	9.489	0.9966
Sartaguda	0.980	0.719	0.9656	1.513	2.618	0.9956
Tafalla	1.211	0.746	0.9396	1.630	3.368	0.9827
Trinidad	2.023	0.520	0.9660	2.164	8.466	0.9945
Ujué	1.854	0.603	0.9205	1.911	6.441	0.9963
Yesa	1.009	0.665	0.9887	1.489	2.854	0.9978
Yugo	1.720	0.636	0.9384	1.982	5.943	0.9973

using a nonlinear regression. The result appears in Table 2.

4. ANALYSIS OF RESULTS

To start with let us consider the statistical analysis results, that can be seen in Table 3. There are nine stations with wind speed values over 5 m s^{-1} . Only three of them are in the southern half of the study area. The highest one corresponds to El Perdón, with a mean speed of 8.6 m s^{-1} . These stations have also the highest median, standard deviation and maximum

wind speed. The minimum values are similar for all of them.

If we observe the values for the scale parameter c of Weibull, it seems clear that it is near to the mean speed. This is true because the Weibull distribution is a reasonable fit to the data. At this point, it is necessary to study how the models conform to our data. We will use the traditional approach for the Weibull distribution consisting of plotting the data in the form of $\ln(-\ln(1-\text{cumulative probability}))$ against $\ln v_i$. From the R^2 correlation coefficient, obtained from the least squares fit of the line,

Table 3. Samples analysis results

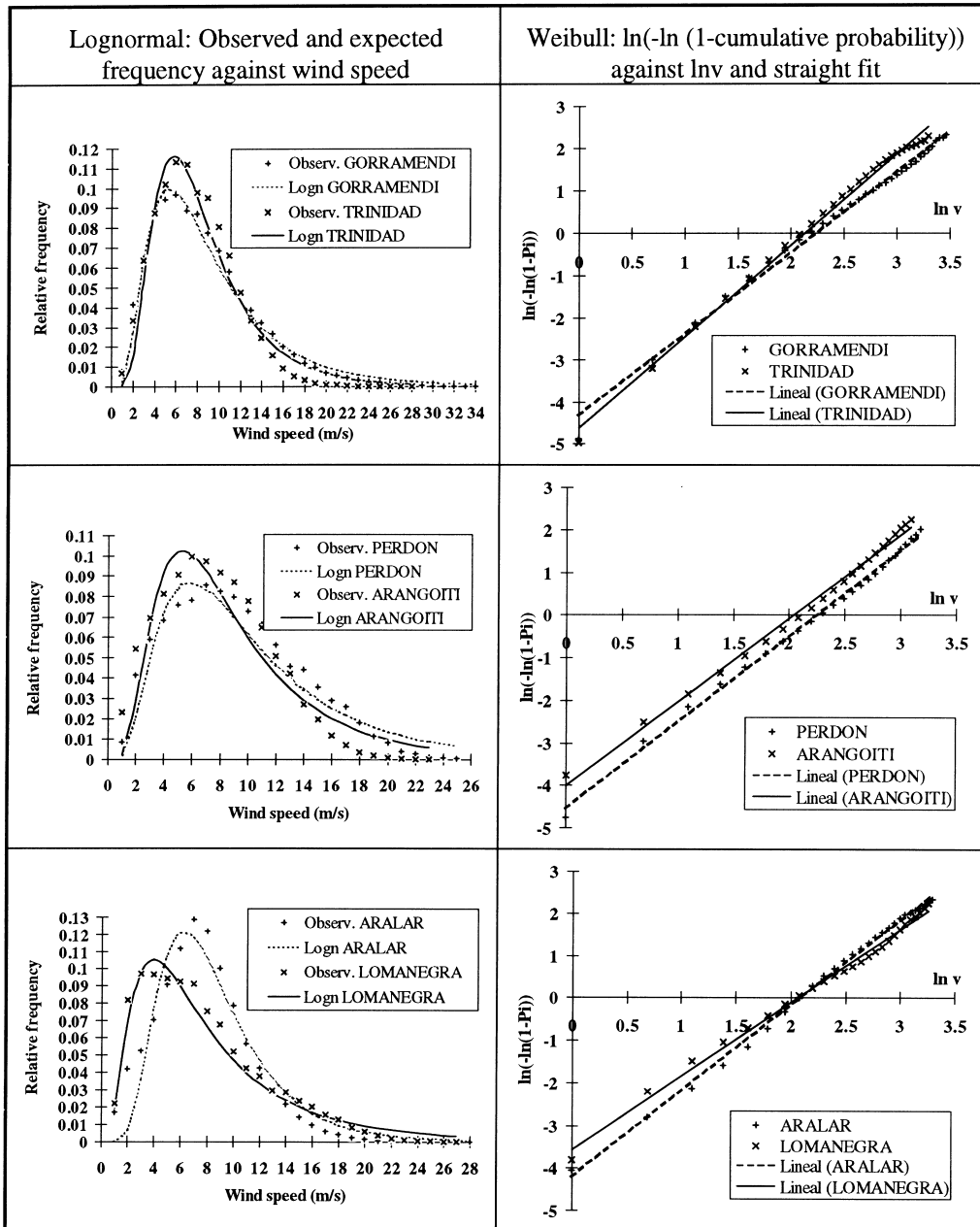
Station	Sample size	Average	Median	Standard deviation	Variance	Maximum	Minimum	Kurtosis	Skewness
Aguilar	34957	4.3965	4.1	2.3728	5.6299	16.3	0	0.5436	0.7228
Aoiz	34948	2.2669	2.1	1.5274	2.333	13.3	0	0.5495	0.7141
Aralar	31978	7.2179	7	3.4787	12.1014	27.3	0	0.601	0.574
Arangoiti	24670	7.197	6.9	3.7523	14.0796	22.5	0	-0.3442	0.3921
Cadreita	8015	2.6916	2.2	2.1504	4.6243	11.6	0	-0.0395	0.8305
Carcastillo	34843	2.9688	2.6	1.8848	3.5526	13.5	0.1	0.4434	0.8736
Carrascal	34207	6.3724	5.9	3.584	12.8449	23.1	0	-0.1745	0.5893
Estella	7265	1.7643	1.5	1.1309	1.279	10.9	0.1	3.6331	1.4135
Etxarri	8433	2.4545	2.2	1.8935	3.5853	1.6	0	5.4277	1.791
Gorramendi	24694	8.0843	7.3	4.6443	21.5699	33.2	0	0.9493	0.9595
Ilundain	15615	1.613	1.3	1.4152	2.0029	9.7	0	3.8968	1.7143
Loma Negra	33886	7.1038	6.2	4.4968	20.2212	26.5	0	0.2348	0.8574
Noain	8648	3.2091	2.8	2.5475	6.4898	13.3	0	-0.6802	0.5261
El Perdón	17518	8.5927	8.1	4.5584	20.7787	24.9	0.2	-0.3292	0.4995
Sartaguda	15411	2.4722	2.2	1.6702	2.7895	12	0	0.5237	0.8292
Tafalla	8748	3.2276	2.9	2.0751	4.306	11.2	0.1	-0.3238	0.6349
Trinidad	33232	7.2587	6.9	3.4715	12.0511	27.4	0.1	0.3119	0.5854
Ujué	24711	5.9545	5.7	3.1872	10.158	19.6	0	-0.0497	0.502
Yesa	8714	2.6528	2.2	1.8102	3.2769	11.3	0.1	1.007	1.0579
Yugo	27881	6.0522	5.4	4.1248	17.0142	31.8	0	4.4106	1.8163

we can judge the suitability of the distribution. With regard to the Lognormal distribution, the procedure obtains least squares estimates of the parameters in the nonlinear regression model and also the R^2 coefficient, using a search algorithm in an attempt to minimize the residual sum of squares. All the parameters and R^2 coefficients can be seen in Table 2. Figure 2

includes plots for six sites, chosen because they have mean wind speeds over 7 m s^{-1} .

4.1. An example: El Perdón wind farm

The first wind farm in Navarre was installed near to Pamplona. It was divided in two phases. The first phase started to run in December 1994 and consisted of six 500 kW wind generators



Obsv.: Observed frequency
Logn : Expected frequency for Lognormal

+ , x : Observed frequency in the form of reduced variate (y) against $\ln v$
Lineal : Least squares straight line fit

Fig. 2. Suitability of Weibull and Lognormal distributions for the stations with mean speed $> 7 \text{ ms}^{-1}$

with an installed power of 3 MW. In March 1996, the second phase started to run with other 34 wind generators. The total installed power is 20 MW. During 1995 the first phase showed a production of 11.66 GWh. The estimation was of 11.85 GWh (using a Weibull distribution with $k=1.881$ and $c=9.922$), turning out an efficiency of 98.41% (Jiménez, 1996).

5. CONCLUSIONS

Only for the stations of Estella and Ilundain, with the lowest mean speed (less than 2 m s^{-1}), the Lognormal distribution fits the data better than the Weibull distribution. For the rest of the stations, this analysis leads to the conclusion that the Weibull distribution is the best distribution and provides a very useful model to estimate the potential wind energy.

Our future aim consists of starting from a statistically ratified model, seeing whether we can obtain a relation between parameters and the orography of the community. The goal is the creation of a contour map plotted for the

parameters obtained of the best-fit function. Those maps may be a good base for a rough forecast of the parameters at locations where the wind data are not available.

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